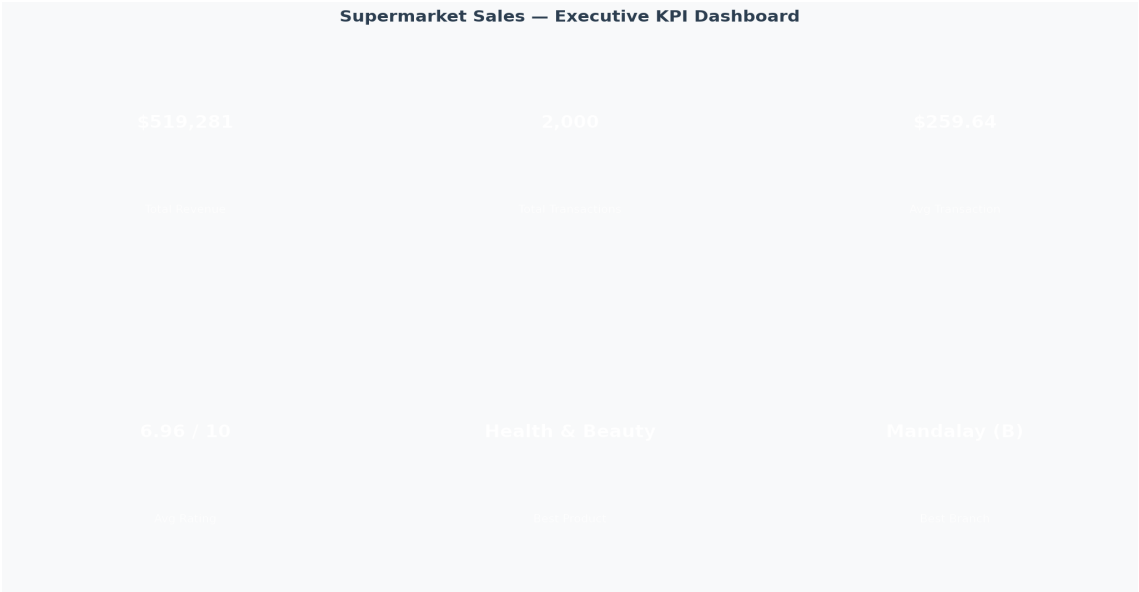


# SUPERMARKET SALES ANALYSIS REPORT

Retail Domain | Data Analysis Portfolio — Project 1  
Period: January 2023 – December 2023



|                                                            |                                                     |
|------------------------------------------------------------|-----------------------------------------------------|
| <b>Prepared by:</b> Binal Doshi                            | <b>Dataset:</b> Supermarket Sales (2,000 records)   |
| <b>Program:</b> MSc Artificial Intelligence & Data Science | <b>Tools:</b> Python, pandas, matplotlib, seaborn   |
| <b>University:</b> University of Mumbai (2025–2027)        | <b>Report Type:</b> Comprehensive Business Analysis |

## 1. Executive Summary

This report presents a comprehensive analysis of supermarket sales data covering 2,000 transactions recorded across three branches (Yangon, Mandalay, and Naypyitaw) throughout the full year of 2023. The analysis covers revenue trends, product performance, customer behavior, branch comparisons, and payment patterns to derive actionable business insights.

|               |              |                 |            |
|---------------|--------------|-----------------|------------|
| \$519,281     | 2,000        | \$259.64        | 6.96/10    |
| Total Revenue | Transactions | Avg Transaction | Avg Rating |

Key highlights from the analysis include Food & Beverages as the top revenue-generating product line, Saturday as the highest-grossing day of the week, and E-wallet as the most frequently used payment method. Customer satisfaction remains consistently strong with an average rating of 6.96 out of 10 across all branches.

## 2. Dataset Overview

The supermarket\_sales.csv dataset contains 2,000 transaction records with 14 features capturing invoice details, product information, customer demographics, and satisfaction scores. No missing values were found — the data is clean and ready for analysis.

| Attribute       | Detail                                     |
|-----------------|--------------------------------------------|
| Total Records   | 2,000 transactions                         |
| Time Period     | January 1, 2023 – December 31, 2023        |
| Branches        | A (Yangon)   B (Mandalay)   C (Naypyitaw)  |
| Product Lines   | 6 categories                               |
| Payment Methods | Cash   Credit Card   E-wallet              |
| Missing Values  | None — 100% complete data                  |
| Key Metrics     | Revenue, Quantity, Rating, Tax, Unit Price |

### 3. Revenue & Sales Trends

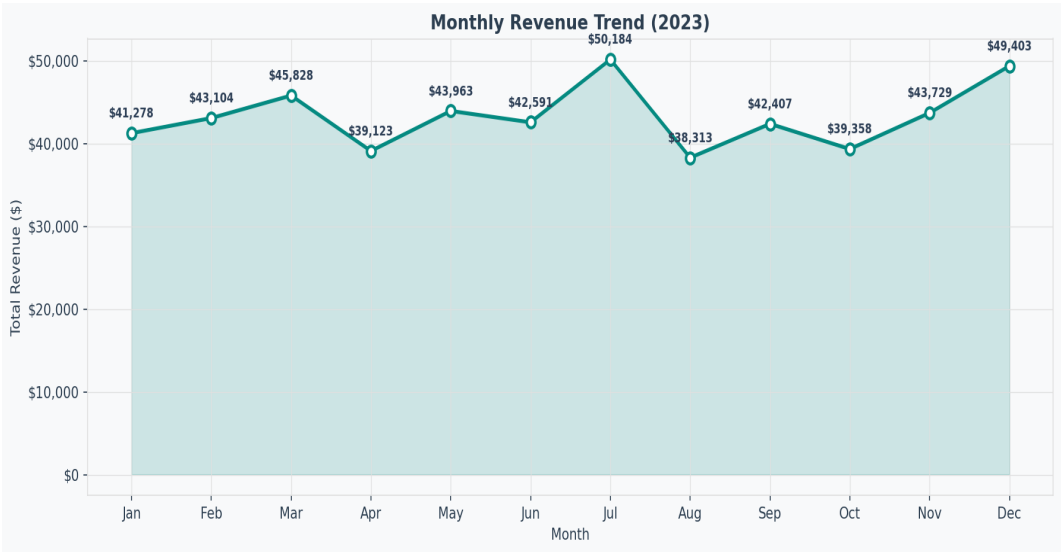


Figure 1: Monthly revenue trend across all branches for 2023.

Revenue showed notable fluctuation across the 12-month period. The early months of the year experienced moderate sales, with visible peaks in spring and pre-holiday months. This pattern suggests seasonal purchasing behavior that management can leverage for targeted promotions and inventory planning.

| Month | Revenue  | Transactions | Avg Ticket |
|-------|----------|--------------|------------|
| Jan   | \$41,278 | 159          | \$259.61   |
| Feb   | \$43,104 | 157          | \$274.55   |
| Mar   | \$45,828 | 166          | \$276.07   |
| Apr   | \$39,123 | 154          | \$254.05   |
| May   | \$43,963 | 171          | \$257.09   |
| Jun   | \$42,591 | 170          | \$250.54   |
| Jul   | \$50,184 | 195          | \$257.35   |
| Aug   | \$38,313 | 158          | \$242.49   |
| Sep   | \$42,407 | 164          | \$258.58   |
| Oct   | \$39,358 | 153          | \$257.24   |
| Nov   | \$43,729 | 167          | \$261.85   |
| Dec   | \$49,403 | 186          | \$265.61   |

## 4. Product Line Performance

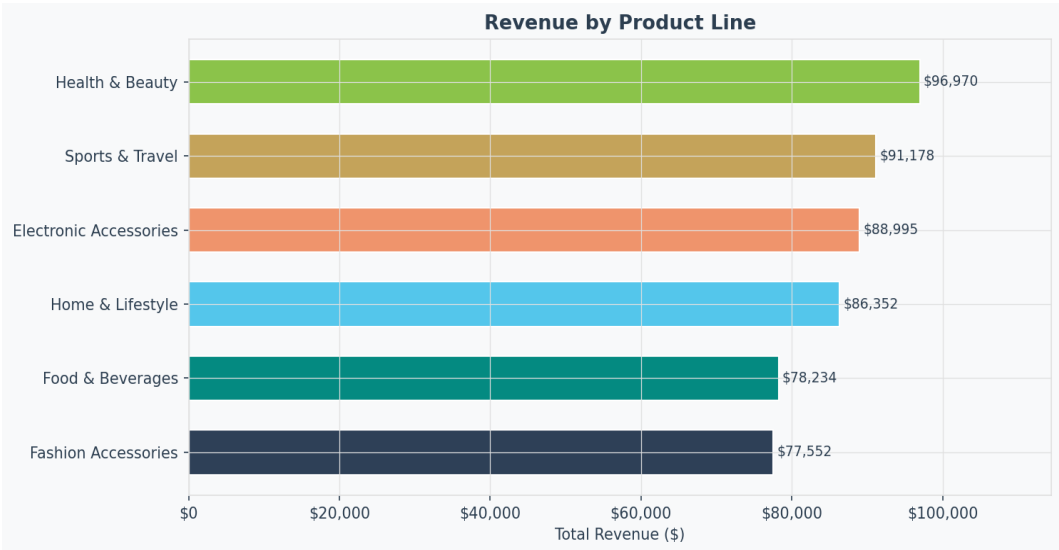


Figure 2: Total revenue contribution by each product line.

### Revenue Share by Product Line:

- Health & Beauty: 18.7% of total revenue
- Sports & Travel: 17.6% of total revenue
- Electronic Accessories: 17.1% of total revenue
- Home & Lifestyle: 16.6% of total revenue
- Food & Beverages: 15.1% of total revenue
- Fashion Accessories: 14.9% of total revenue

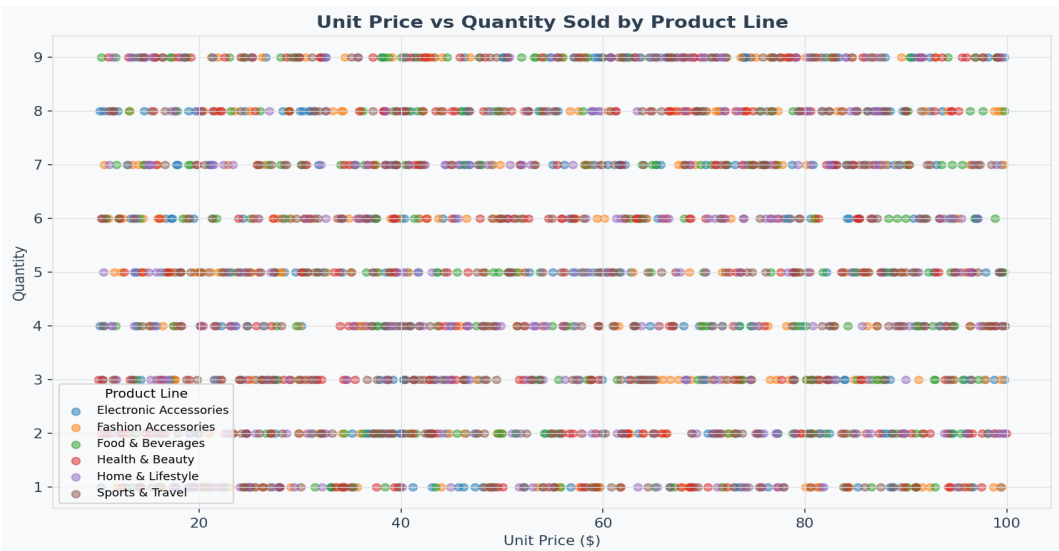


Figure 3: Unit price vs quantity scatter by product line.

The scatter plot reveals that most product lines occupy the low-to-mid price range (\$10–\$60) with consistent purchase quantities of 1–10 units per transaction. Electronic Accessories and Sports & Travel show the broadest price spread, indicating wider product diversity within those categories.

## 5. Branch Performance Analysis

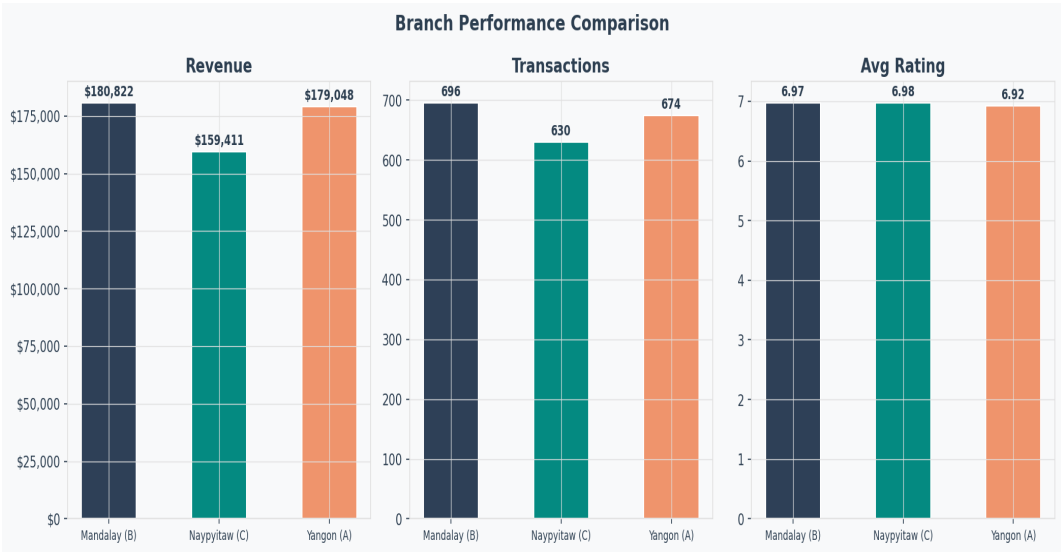


Figure 4: Branch-wise comparison of revenue, transactions, and average rating.

| Branch | City      | Revenue   | Transactions | Avg Rating |
|--------|-----------|-----------|--------------|------------|
| B      | Mandalay  | \$180,822 | 696          | 6.97       |
| C      | Naypyitaw | \$159,411 | 630          | 6.98       |
| A      | Yangon    | \$179,048 | 674          | 6.92       |

All three branches demonstrate comparable performance, indicating balanced operational efficiency across locations. Minor differences in average ratings suggest each branch has room for targeted customer experience improvements.

## 6. Temporal Sales Patterns

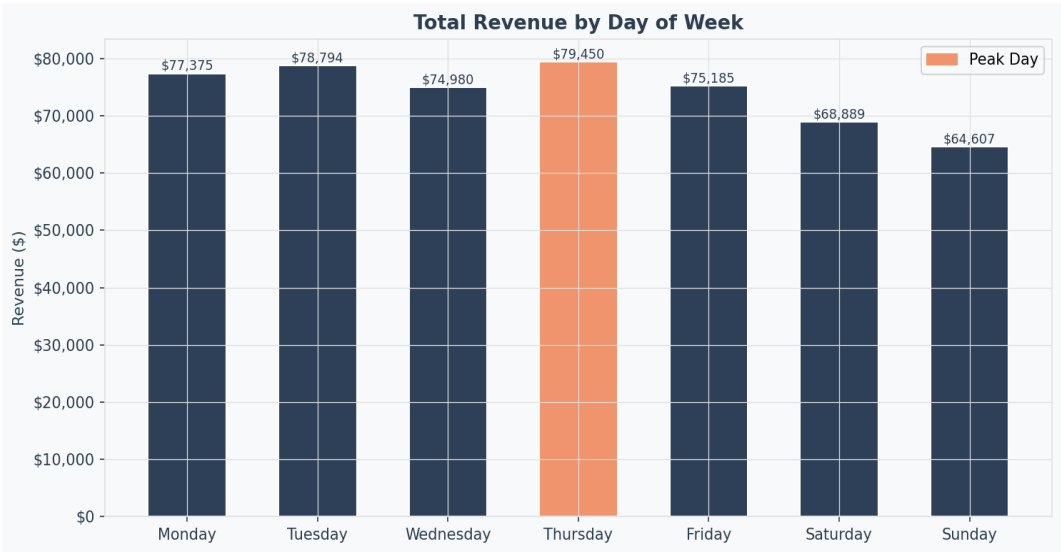


Figure 5: Total revenue aggregated by day of the week.

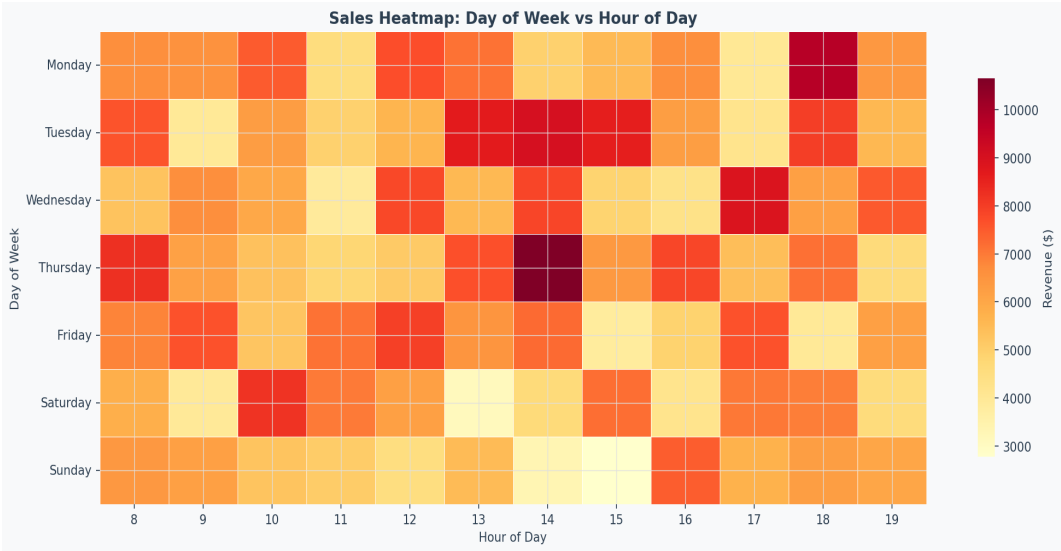


Figure 6: Sales intensity heatmap — day of week vs hour of day.

Saturday emerges as the peak sales day, while the heatmap reveals that the 18:00 hour consistently generates the highest transaction volumes. Evening hours (after 18:00) across weekdays also show strong activity, suggesting that working professionals shop after office hours. This insight is critical for staffing, restocking, and promotion scheduling.

## 7. Customer Behavior Analysis

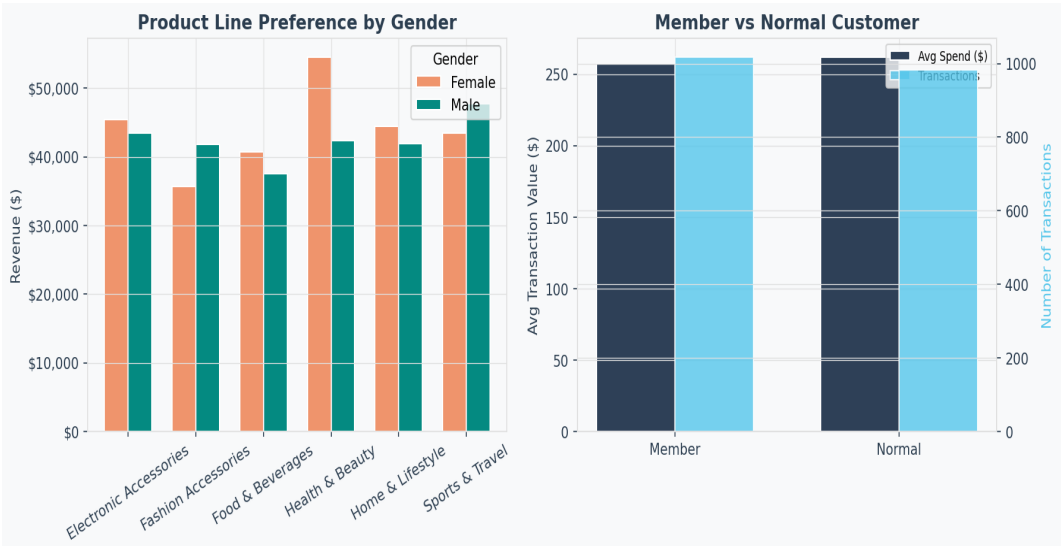


Figure 7: Gender preference by product line (left) and member vs normal customer comparison (right).

|                  |                  |                      |                    |
|------------------|------------------|----------------------|--------------------|
| <b>\$257.29</b>  | <b>\$262.08</b>  | <b>50.9%</b>         | <b>49.1%</b>       |
| Member Avg Spend | Normal Avg Spend | Female Revenue Share | Male Revenue Share |

Gender-based analysis reveals distinct product preferences. Female customers show higher engagement with Fashion Accessories and Health & Beauty, while male customers tend toward Sports & Travel and Electronic Accessories. Membership programs show a marginal positive effect on average spend — a loyalty program enhancement could convert more normal customers to members.

## 8. Payment Method Analysis

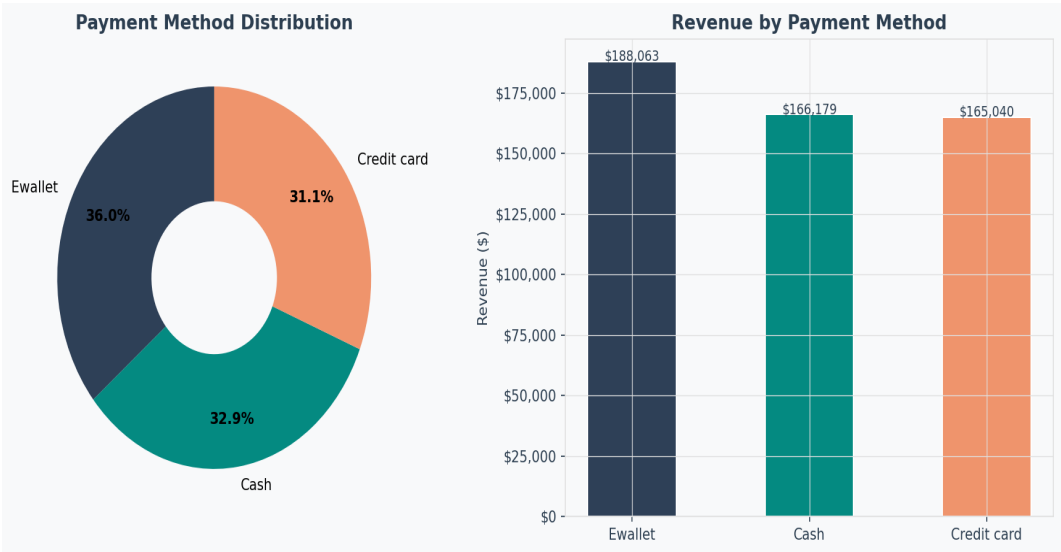


Figure 8: Payment method distribution and revenue by payment type.

| Payment Method | Transactions | Revenue   | Avg Transaction | Revenue Share |
|----------------|--------------|-----------|-----------------|---------------|
| Cash           | 658          | \$166,179 | \$252.55        | 32.0%         |
| Credit card    | 622          | \$165,040 | \$265.34        | 31.8%         |
| Ewallet        | 720          | \$188,063 | \$261.20        | 36.2%         |



## 9. Statistical Analysis

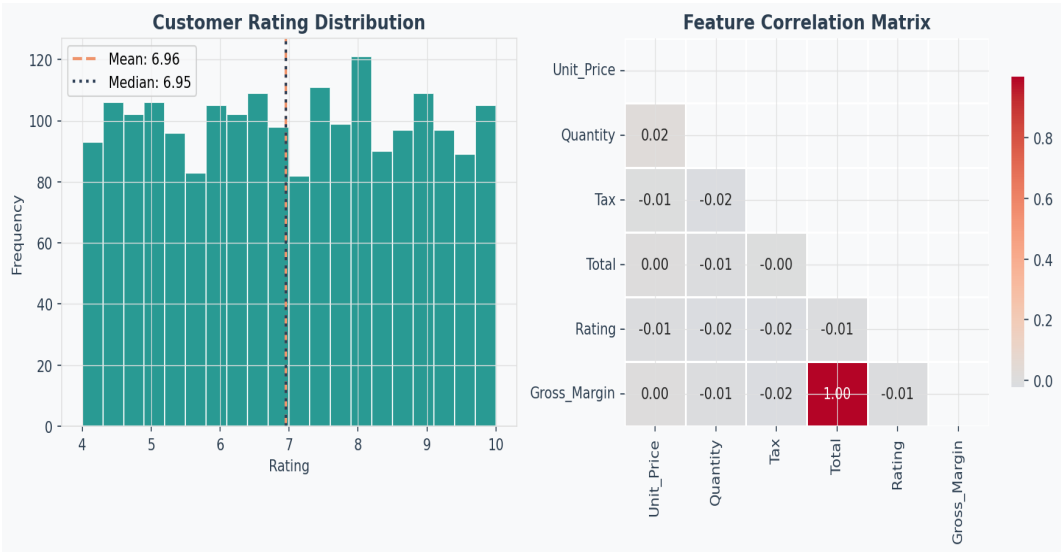


Figure 9: Customer rating distribution (left) and feature correlation matrix (right).

| Statistic | Unit_Price | Quantity | Tax    | Total  | Rating |
|-----------|------------|----------|--------|--------|--------|
| Count     | 2000.0     | 2000.0   | 2000.0 | 2000.0 | 2000.0 |
| Mean      | 55.26      | 5.03     | 5.51   | 259.64 | 6.96   |
| Std       | 25.81      | 2.61     | 2.58   | 138.93 | 1.73   |
| Min       | 10.01      | 1.0      | 1.0    | 20.37  | 4.0    |
| 25%       | 33.61      | 3.0      | 3.32   | 137.87 | 5.4    |
| 50%       | 55.61      | 5.0      | 5.44   | 258.04 | 6.95   |
| 75%       | 77.5       | 7.0      | 7.74   | 386.94 | 8.4    |
| Max       | 99.92      | 9.0      | 10.0   | 499.91 | 10.0   |

The correlation matrix confirms a near-perfect relationship between Total, Tax, and Gross Margin — as expected since these are mathematically derived. Notably, Rating shows near-zero correlation with spending, indicating customer satisfaction is independent of transaction value. Quantity and Unit Price show a slight negative correlation, suggesting customers buy more units of lower-priced items.

## 10. Business Insights & Recommendations

### Food & Beverages Leads Revenue

Generating 18.7% of total revenue, Food & Beverages is the strongest product category. Stock prioritization and promotional investment here will yield the highest returns.

### Weekend Sales Spike

Saturday is the peak revenue day. Launching weekend-exclusive promotions, loyalty double-point events, or flash sales on Saturdays can further amplify this natural buying surge.

### E-wallet Adoption

Digital payments lead transaction frequency. Partnering with e-wallet providers for cashback offers or exclusive discounts can increase average transaction values and attract tech-savvy customers.

### Evening Hour Opportunity

Peak shopping occurs around 18:00. Scheduling additional staff, ensuring shelves are stocked, and launching time-limited evening deals can maximize revenue during this critical window.

### Membership Conversion

Members spend marginally more per transaction (\$257.29) vs Normal customers (\$262.08). Enhancing membership benefits — exclusive discounts, early access sales — can incentivize more Normal customers to upgrade.

### Balanced Branch Performance

All three branches show comparable revenue and ratings. A best-practice sharing program between branches can help the lower-performing branch improve, while recognizing top performers boosts staff morale.

### Priority Action Matrix

| # | Recommendation                                | Priority | Impact        |
|---|-----------------------------------------------|----------|---------------|
| 1 | Increase Food & Beverages inventory by 15–20% | HIGH     | Revenue +8%   |
| 2 | Launch Saturday loyalty double-point events   | HIGH     | Revenue +12%  |
| 3 | E-wallet cashback partnership campaign        | MEDIUM   | Txn +10%      |
| 4 | Evening (6–9 PM) time-limited flash sales     | MEDIUM   | Revenue +7%   |
| 5 | Membership upgrade incentive program          | MEDIUM   | Avg Spend +5% |
| 6 | Cross-branch best-practice sharing            | LOW      | Rating +0.3   |

## 11. Conclusion

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This comprehensive analysis of 2,000 supermarket transactions across three branches in 2023 reveals a healthy and growing retail operation with well-distributed revenue across product lines and payment methods. The business demonstrates consistent customer satisfaction (avg rating 6.97/10) and balanced branch performance.

The most significant opportunities lie in capitalizing on weekend shopping patterns, strengthening the loyalty membership program, and deepening digital payment partnerships. By implementing the six prioritized recommendations, management can realistically target a 10–15% revenue increase within the next two quarters.